Yujie Gao | New York, NY | ripapply17@gmail.com | Website

EDUCATION

Cornell Tech, Cornell University New York, NY
MEng in Operations Research and Information Engineering, Merit Scholarship May 2026
University of California, Berkeley Berkeley, CA
BA in Data Science May 2025

PROFESSIONAL EXPERIENCE

Goodyear Tire, Data Scientist Intern **May 2024 - August 2024**

- Engineered a Python/SQL ETL pipeline in Snowflake to ingest and clean 50,000+ monthly sales records, implementing feature specific imputation strategies including forward-fill, and segment-level aggregation, reducing reporting latency by 10 hours/month.
- Processed 10M+ raw vehicle registration records from the CAAM database and applied K-Means geospatial clustering algorithms to segment over 300 regional markets, successfully identifying 5 high-yield distribution nodes for optimized logistics routing.
- Developed a regularized Ridge regression model for regional demand forecasting, integrating seasonal and macroeconomic indicators to improve out-of-sample R^2 from 0.79 to 0.82 and reduce overall forecast variance by 15% across operating regions.
- Built interactive Tableau diagnostic dashboards to continuously operationalize core supply chain KPIs, enabling regional managers across 30+ provinces to proactively resolve inventory bottlenecks based on real-time Actuals versus Forecast variance analysis.

Baidu Internet Service, Data Scientist Intern (Ads Ecosystem) **May 2023- August 2023**

- Managed comprehensive A/B testing frameworks for dynamic ad creative ranking, utilizing Baidu's internal experimentation platforms to evaluate p-values and tight confidence intervals to ensure the safety and efficacy of high-traffic feature rollouts.
- Investigated a 5% relative Conversion Rate drop within a specific iOS cohort via multidimensional user-path SQL segmentations, collaborating closely with product engineering to deploy a targeted landing-page fix that restored baseline conversion stability.
- Created real-time tracking dashboards to closely monitor Ads platform metrics, translating petabyte-scale unstructured behavior logs into diagnostic KPIs: eCPM, CTR, and Hide-rate to optimally balance monetization targets with long-term user retention.
- Optimized distributed SparkSQL and Hive queries to efficiently extract actionable business insights from unstructured user activity datasets, significantly reducing overall data processing latency and accelerating the delivery of weekly ecosystem health reviews.

PROJECTS

Bird Audio Classification & Sound Event Detection (BirdClef 2026) | PyTorch, ONNX, CNN, Spectrum | Deep Dive **2026**
Developed an end-to-end bioacoustics deep learning pipeline to accurately identify avian vocalizations within continuous, highly noisy field recordings, engineering the system to operate seamlessly under strict, low-latency computational deployment constraints.

- Engineered a scalable audio preprocessing pipeline utilizing TorchAudio to transform raw 1D audio into optimized 2D Mel-spectrogram, successfully training robust CNN-based neural network backbones for complex multi-label sound event detection.
- Mitigated severe long-tailed class imbalance and out-of-distribution domain shifts by designing EDA-driven data augmentation strategies, including MixUp and CutMix, to enhance overall model generalization against complex, real-world background noise.
- Leveraged Knowledge Distillation to transfer continuous logit predictions from a complex ensemble of deep teacher architectures into a highly lightweight student model, boosting the final classification AUC by over 2% while minimizing parameter bloat.
- Exported production-ready PyTorch model weights to an optimized ONNX Runtime environment and implemented a concurrent multithreaded CPU inference pipeline, accelerating raw execution speeds by 10% to rigorously satisfy low-latency requirements.

Algorithmic Pricing & Dynamic Programming Engine | XGBoost, SK-learn, K-Means, Backward Induction | Deep Dive **2025**
Developed an end-to-end dynamic pricing system that handles real-time revenue optimization by strategically decoupling offline machine learning demand estimation models from a highly responsive online finite-horizon optimization solver.

- Built a robust finite-horizon dynamic programming pricing solver utilizing K-Means state discretization and static precomputed policy tables, structurally guaranteeing instantaneous O(1) online pricing decisions to navigate a strict 0.5-second SLA constraint.
- Developed an advanced XGBoost purchase-probability model in Scikit-learn to support rigorous pricing optimization, outperforming baseline models by 39.1% in total simulated profit across a comprehensive 50K-customer holdout test set.
- Deployed a dynamic adaptive feedback loop into a live competitive tournament environment, algorithmically adjusting pricing strategies based strictly on recent transaction win-rates and inventory depletion speeds to secure a top 10 of competing teams.

Causal Inference & Environmental Health Modeling | GLM, IPW, Statsmodels, GMM, Bootstrap | Deep Dive **2025**
Constructed a causal inference pipeline on observational panel data to estimate the causal impact of ambient air pollution on adult asthma prevalence across California, simultaneously engineering robust spatiotemporal predictive pipelines.

- Built a robust causal inference pipeline using Inverse Propensity Weighting and 1000 bootstrap iterations to simulate a control trial on observational data, isolating Average Treatment Effect of -0.38 on asthma rates while controlling for demographic confounders.
- Engineered a log-transformed, standardized Gaussian GLM that mitigated the overfitting observed in baseline Random Forest models when predicting statewide asthma prevalence, driving a 47000-point reduction in the Akaike Information Criterion.
- Extracted statistical signals by utilizing a Gaussian Mixture Model to mathematically discretize bimodal exposure distributions, and applied permutation importance to eliminate multicollinearity, achieving a RMSE of 15.07 on unseen WA state data.

Technical Skills

- **Languages & Data Infrastructure:** Python, SQL, PySpark, SparkSQL, Snowflake, Git, ETL Data Pipelines.
- **ML & Deep Learning:** PyTorch, Scikit-learn, XGBoost, CNN, Random Forest, K-Means, GLM, GMM, Time Series Forecasting.
- **Experimentation & Causal Inference:** A/B Testing, Causal Inference (IPW), Bootstrapping, PCA, SMOTE, Statistical Modeling.
- **Ops & Analytics:** ONNX Runtime, Edge Deployment, Multithreaded Inference, Tableau, Diagnostic Dashboards, KPI Tracking.